Aegivex AI - AI Architecture Document

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1. Overview

This document defines the Artificial Intelligence architecture used in Aegivex AI. The AI system acts as the core intelligence layer of the platform, responsible for understanding user requests, collecting blockchain security data, analyzing risks, and generating clear, actionable recommendations. The architecture is designed to be modular, scalable, and easy to extend with new AI capabilities.

2. AI Objectives

The AI system should:

- Understand natural language queries.
- Analyze blockchain security risks.
- Explain technical information in simple language.
- Generate security recommendations.
- Calculate risk scores.
- Support future multi-agent workflows.

3. AI Workflow

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User Request → Intent Detection → Input Validation → Blockchain Data Collection → Risk Analysis Engine → AI Reasoning → Recommendation Generator → Response Formatter → User Response
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4. AI Components

Component Name	Responsibilities
Component 1: Natural	Understand user prompts, detect user intent, extract wallet

Component Name	Responsibilities
Language Processing	addresses, contract addresses, URLs, and transaction hashes.
Component 2: Security Analysis Engine	Wallet analysis, token analysis, smart contract analysis, website analysis, transaction analysis, threat detection.
Component 3: AI Reasoning Engine	Analyze collected data, identify risks, prioritize threats, generate explanations, produce recommendations.
Component 4: Response Generator	Convert technical findings into simple language, display risk level, provide recommendations, suggest next actions.

5. AI Models

Category	Technology / Framework
Primary Model	OpenAI GPT
Framework	LangChain
Workflow Engine	LangGraph
Future Support	Local LLMs, Fine-Tuned Models, Hybrid AI Systems

6. Prompt Engineering

Every request follows a structured prompt template to ensure consistent and reliable AI responses:

- System Prompt
- User Prompt
- Blockchain Context
- Security Context
- AI Instructions
- Output Format

7. Risk Scoring Engine

Each scan produces a Risk Score from 0 to 100, calculated using blockchain signals, reputation data, AI reasoning, and threat indicators.

Score Range	Risk Level		
0 – 20		21 – 40	**Low Risk**
41 – 60	**Medium Risk**		
61 – 80	**High Risk**		
81 – 100	**Critical Risk**		

8. Supported AI Tasks

- Wallet Risk Assessment
- Token Risk Analysis
- Smart Contract Review
- Website Safety Verification
- Transaction Explanation
- Scam Detection
- Phishing Detection
- AI Security Chat

9. AI Input & Output Format

Accepted AI Inputs	Generated AI Outputs
• Wallet Address • Token Contract • Smart Contract Address • Website URL • Transaction Hash • Natural Language Questions	• Risk Score • Risk Level • Summary • Detailed Explanation • Warning Signs • Recommended Actions • Confidence Score

11. AI Memory

The AI stores memory to improve future recommendations and personalization:

- Previous scans
- User conversations
- Scan history
- Frequently analyzed wallets
- User preferences

12. Error Handling

If AI cannot determine a result:

- Return a confidence warning.
- Explain why information is insufficient.
- Suggest additional inputs.
- Avoid making unsupported conclusions.

13. Security & Privacy

The AI system enforces the following security practices:

- Never expose API keys.
- Protect user data.
- Avoid storing sensitive information unnecessarily.
- Encrypt communication.
- Follow secure prompt handling practices.

14. Future AI Enhancements

- Multi-Agent AI Architecture

- Autonomous Security Agents
- Real-Time Wallet Monitoring
- Predictive Threat Intelligence
- Voice-Based AI Assistant
- AI Notification Engine
- Cross-Chain Risk Analysis

15. AI Architecture Principles

- Intelligent
- Explainable
- Secure
- Scalable
- Modular
- Reliable
- User-Friendly

16. Expected Outcome

The AI architecture enables Aegivex AI to function as a trusted Web3 Security Copilot, delivering fast, accurate, and understandable security insights while providing a scalable foundation for future AI-powered blockchain security services.